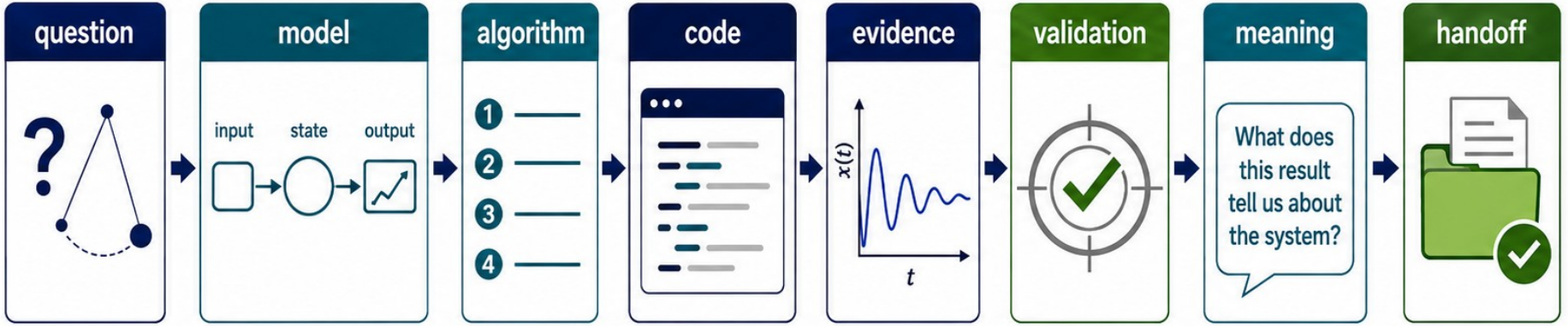
Make One Computational Result Defensible

Week 13 capstone presentation, defence, and handoff

physical question → checked computational result → reproducible handoff



Clarity of evidence matters more than code volume

The Capstone Is Bounded by Design

APPROVED CAPSTONE SCOPE

ONE QUESTION



What physical behaviour are you explaining?

ONE CONTROLLED CHANGE



Change one parameter or assumption

ONE PRINCIPAL OUTPUT



Show one main graph or table

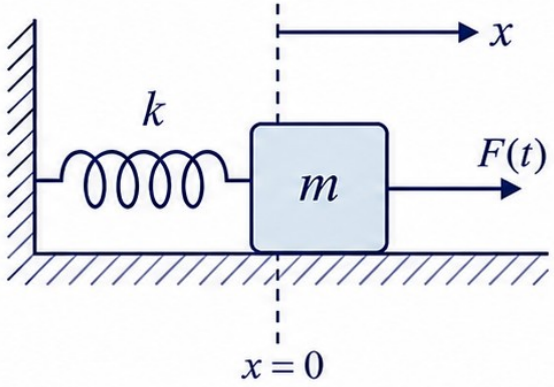
Manageable + assessable

- Use the approved problem space and starter Live Script
- Answer one narrow physical question
- Change one parameter or assumption
- Independent extensions are optional stretch

OPTIONAL STRETCH

Independent extension

Start With the Approved Physical Model



Approved model record

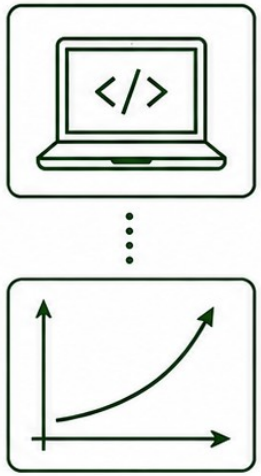
Question
What physical behaviour are we studying?

Assumptions
What is held fixed?

Variables
What changes with time?

Parameters
What values control the model?

Units
Attach units throughout

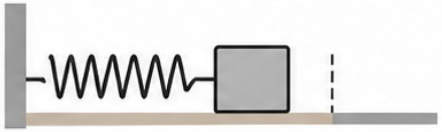


⚠ Do not silently change the approved model

Every important code variable should map to a physical quantity and unit.

Predict the Controlled Change Before Running MATLAB

Baseline state



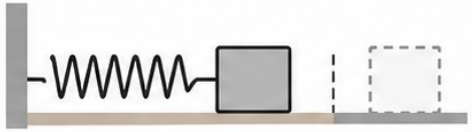
Observe the output

one controlled change

decrease ⟷ increase

predict before run

Modified state



See the final output after running

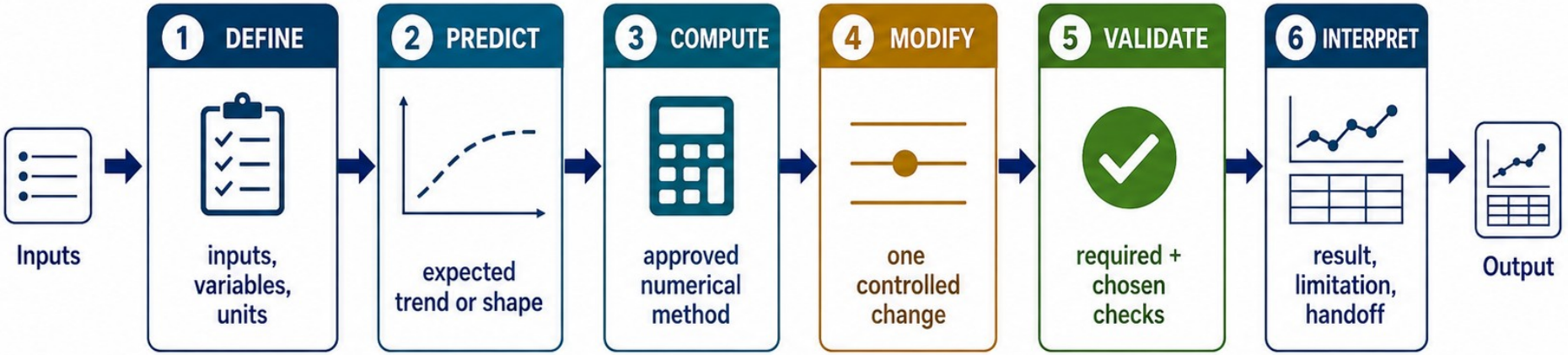
Hold all other inputs fixed

System setup | Environment | Initial conditions | Model structure

- Identify the one parameter or assumption being changed
- State what remains fixed
- Predict the sign, trend, bound, or limiting behaviour
- Record the prediction before showing the final output

My prediction: _____

Complete the Algorithm Before Showing Code

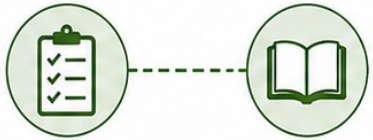


Pseudocode (conceptual)

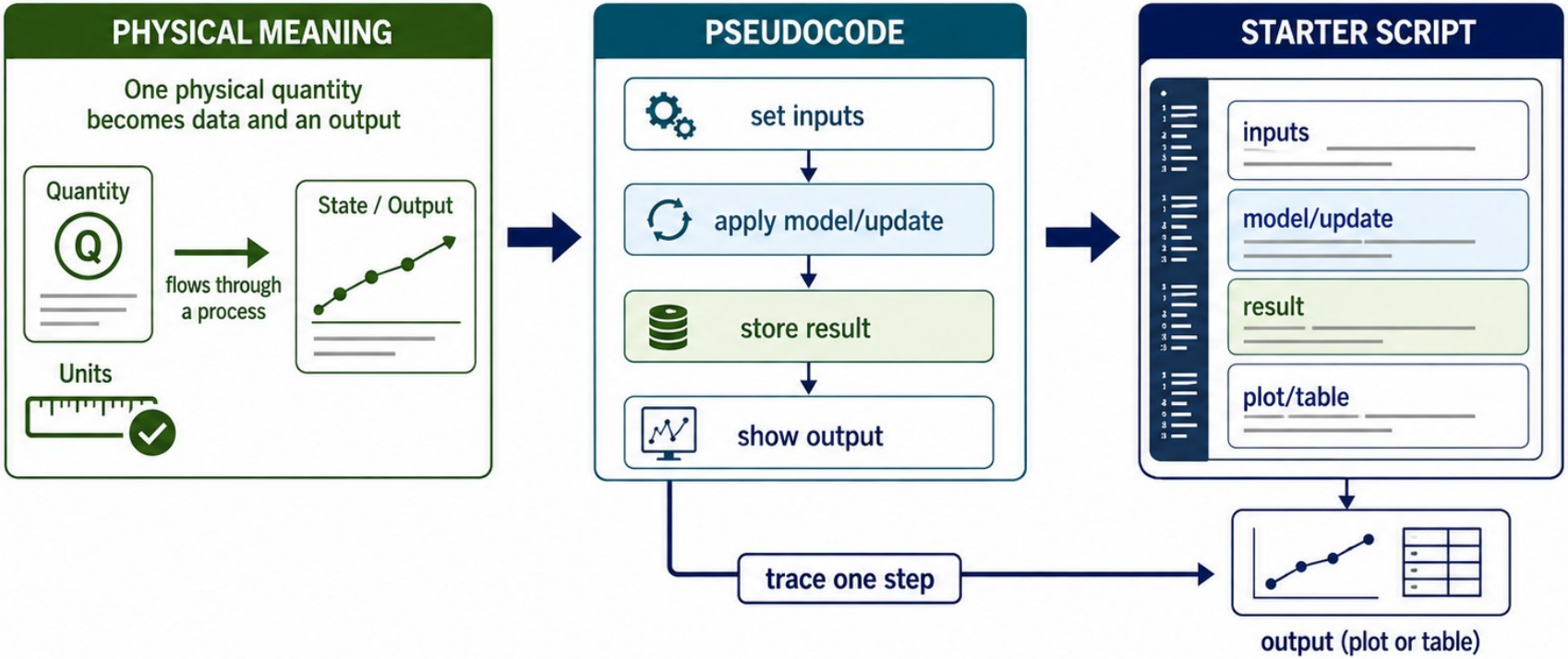
1. read inputs
2. calculate baseline
3. apply one change
4. run checks
5. explain output

Can a teammate predict each code block's job?

Validation checks



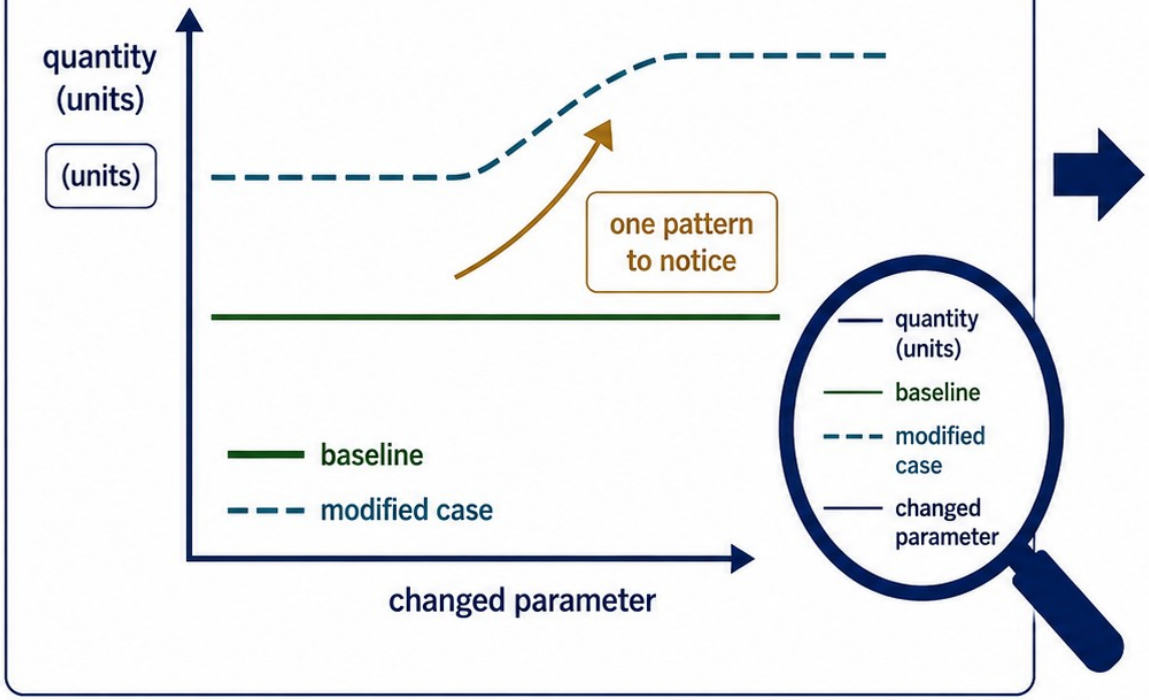
Trace the Starter Script From Physics to Output



- Locate parameter declarations and units
- Trace the model calculation or numerical update
- Find the array, loop, or solver output
- Connect one pseudocode step to one MATLAB line

Show One Principal Output Clearly

PRINCIPAL OUTPUT



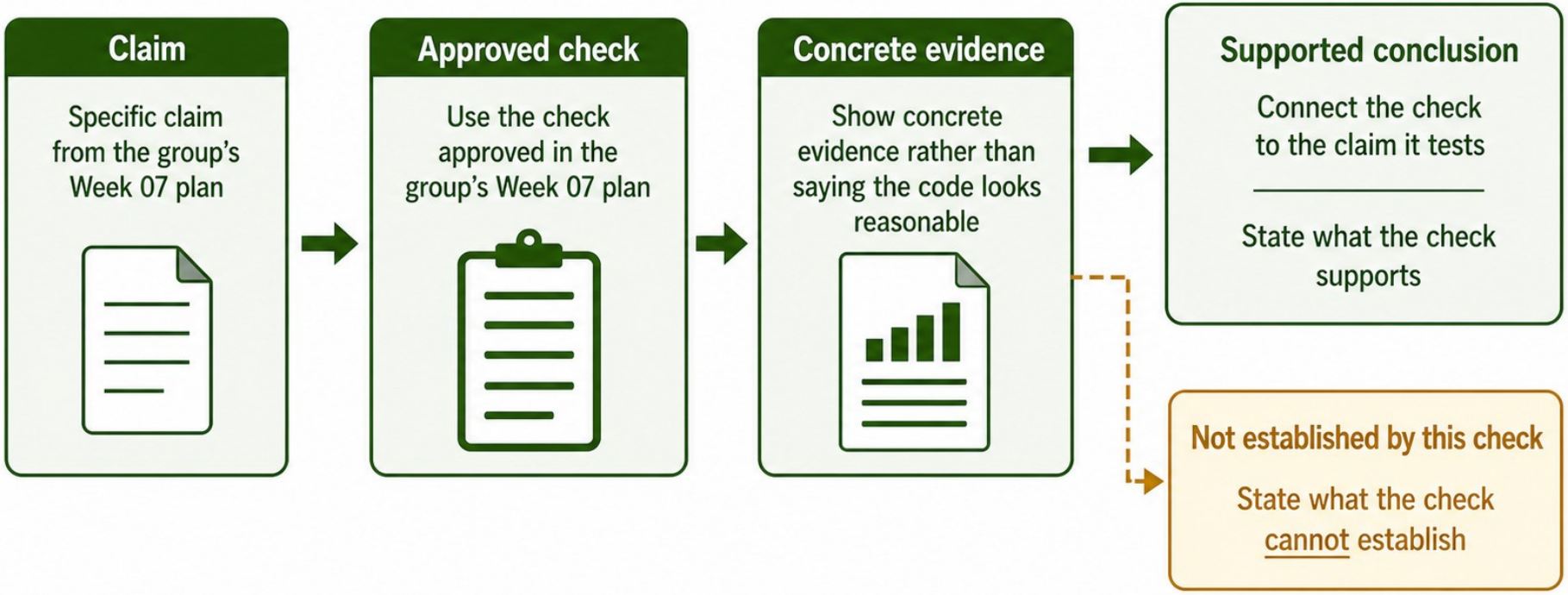
What should the audience notice?

State one clear pattern.

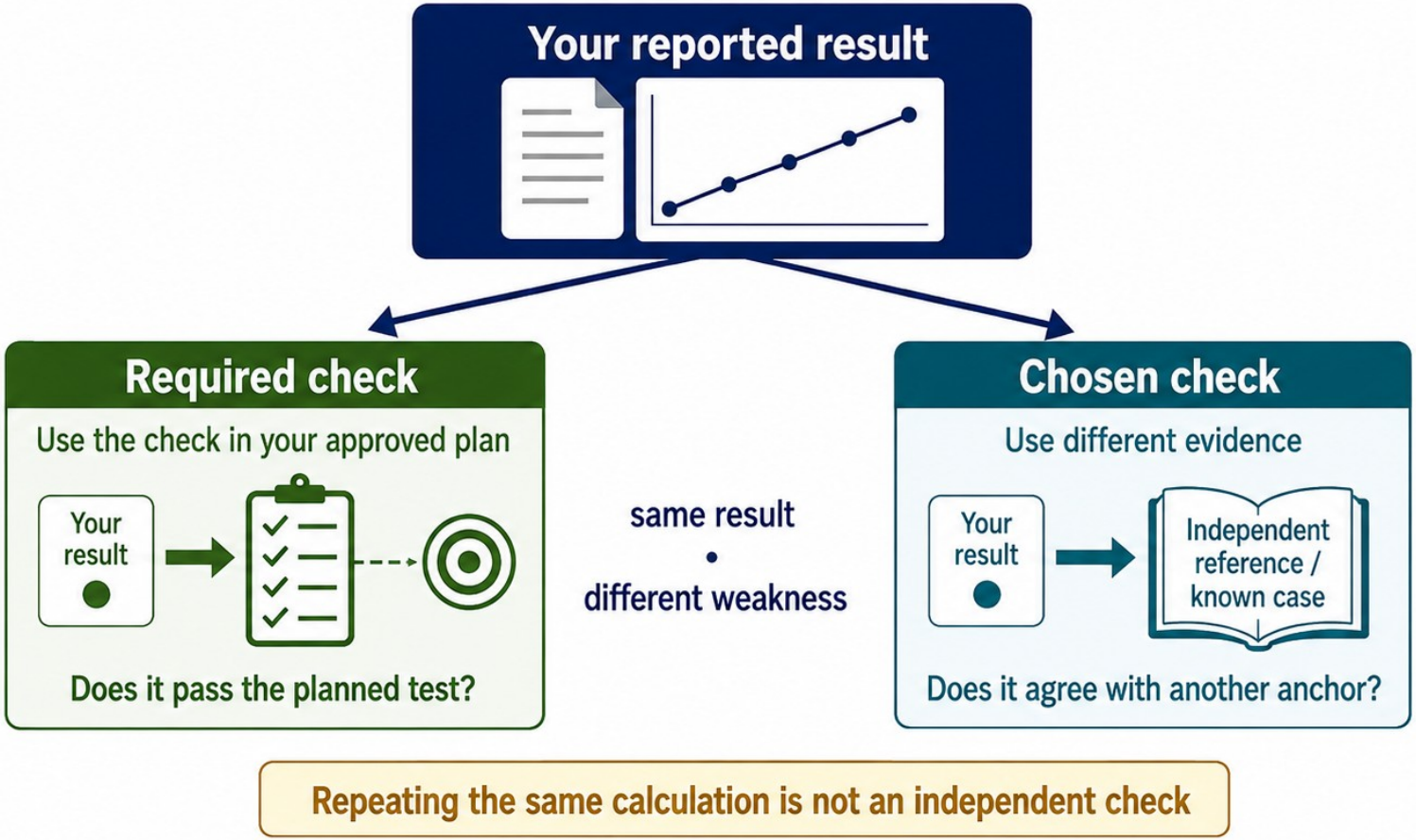
Label: quantities • units • cases

1. Choose one graph or table that answers the physical question.
2. Label quantities, units, cases, and the changed parameter.
3. Distinguish baseline from modified case.
4. State the one pattern the audience should notice.

Apply the Required Validation Check



Add a Second Independent Validation Check

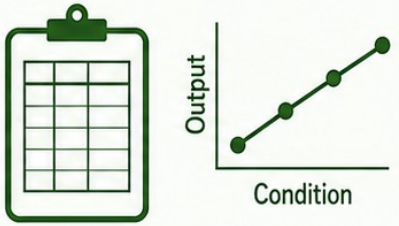


Repeating the same calculation is not an independent check

Interpret the Physics and Name One Limitation

Principal output

What the calculation shows



Refer to the principal output and predicted change

Physical conclusion

What does the result mean?

What changed?

Answer the original physical question in words

One limitation

Where the claim stops



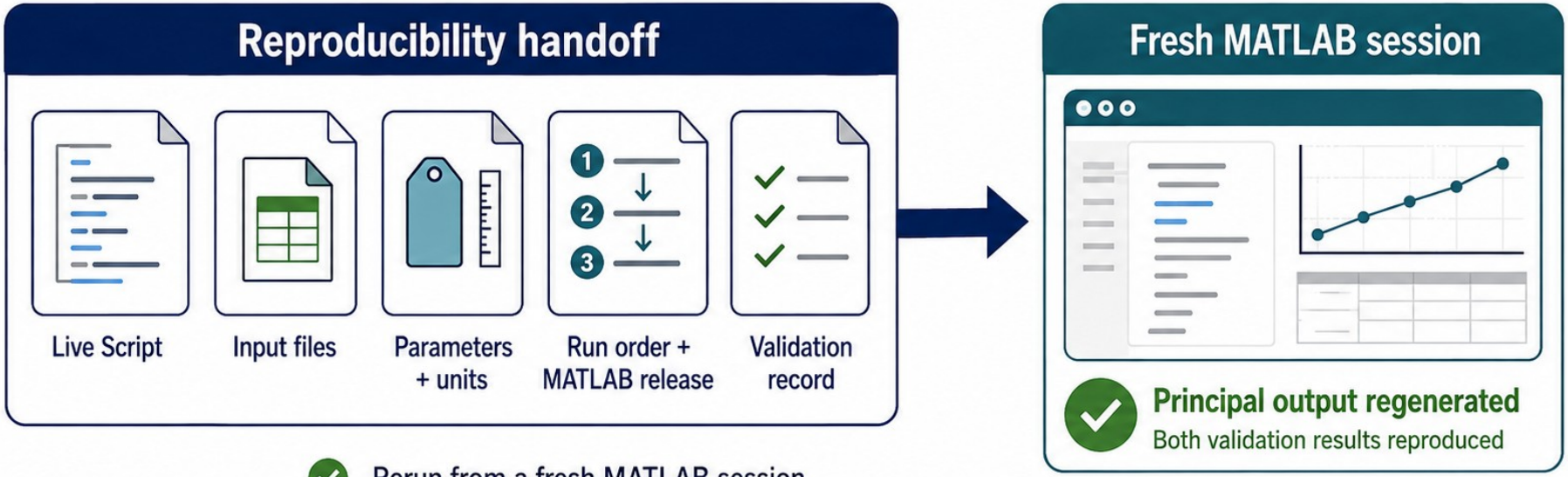
State one limitation even if the code runs

Numerical behaviour
(what the output does)

Model assumption
(what the model assumes)

Separate numerical behaviour from model assumptions

Make the Result Reproducible



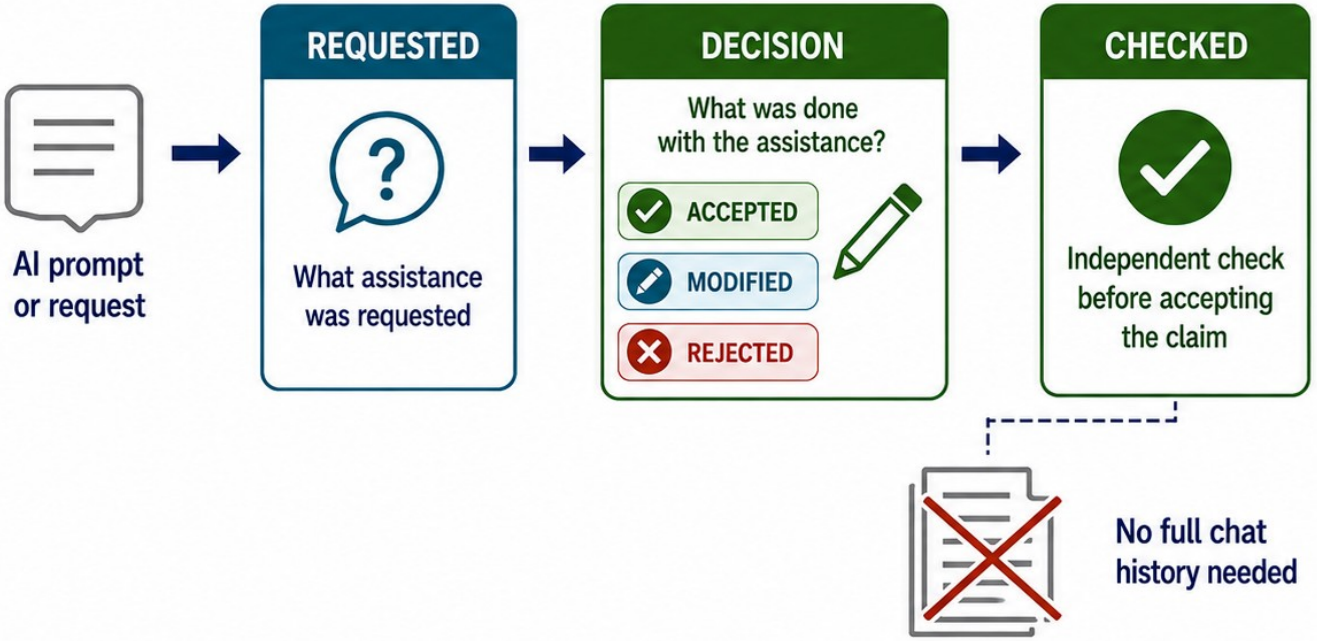
- Rerun from a fresh MATLAB session
- Keep the script and input files together
- Record settings, units, and run order
- Regenerate the output and both checks

Another student should know what to open, run, and check.

Record AI Decisions, Not a Full Chat History

Key Points

- AI is allowed for the capstone when declared
- Record what assistance was requested
- Record what was accepted, modified, or rejected
- State the independent check used before accepting an important claim



Evidence = judgement + independent checking, not transcript length

Prepare for the Individual Defence

Evidence locked

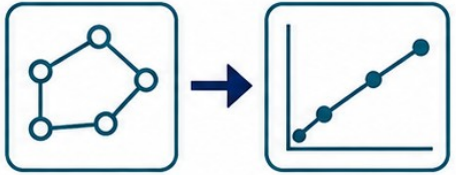
1 Code trace

trace one relevant code section



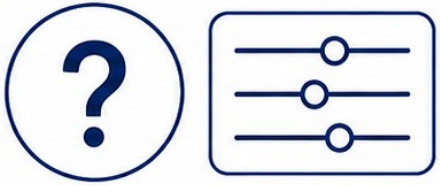
2 Physics explanation

model, method, output, validation



3 Follow-up question

parameter change or validation



Rotation rule 1 → 2 → 3

Member 1 → Member 2 → Member 3

Each member gets a first defence before any second entry.

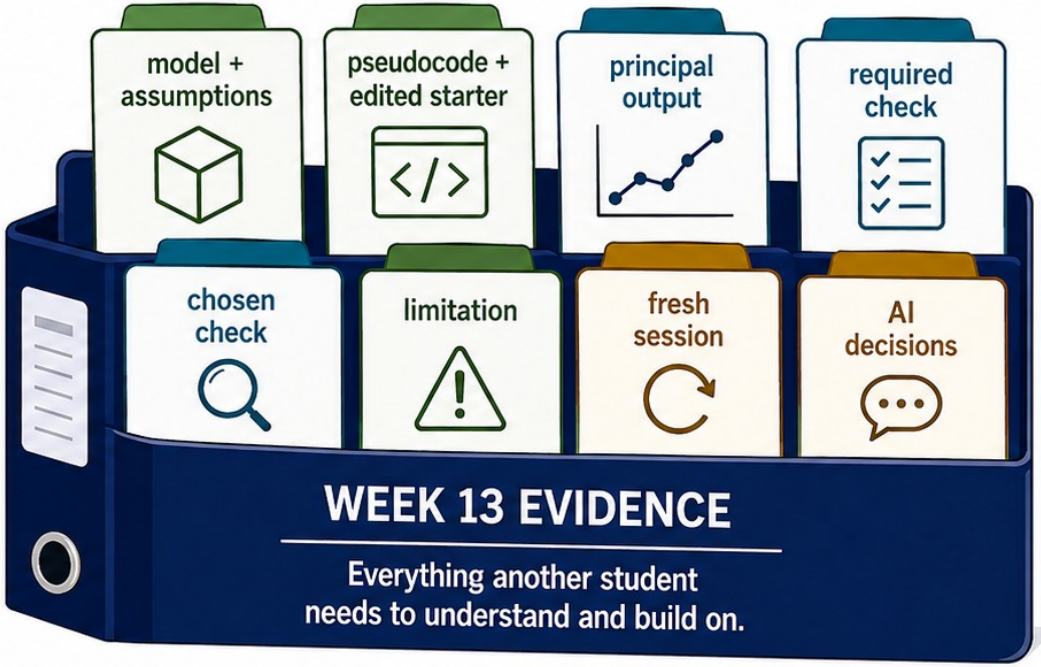
Absence = pending; arrange a make-up.

Defend the evidence, not a blank-page program.

Handoff the Evidence Another Student Can Use

Before you hand off, include:

- ☑ Approved model, assumptions, pseudocode, and edited starter
- ☑ Principal output plus required and chosen checks
- ☑ Physical conclusion, limitation, fresh-session record, and AI decisions
- ☑ Week 13 completes the milestone; Week 14 is recovery only if needed





What did we change?

How do we know?

What remains limited?